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Supplementary information

Atmospheric CO_2 control of spontaneous millennial-scale ice age climate oscillations

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Supplementary Information

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The simple conceptual model

We formulate a simple reduced dimensional dynamical system model of thermohaline oscillations in the glacial Atlantic Ocean in order to produce a simple emulator for the behaviour observed in the comprehensive model. We implement a simplified box model representation of the Atlantic Ocean that will ultimately be generalized to represent a southern box and a northern box. The formulation follows the class of "Stommel type" [1] models that are driven by meridional pressure gradients [2]. The model resolves meridional density gradients calculated from a linear equation of state for seawater (LEOS) which depends on both temperature and salinity (e.g. thermohaline gradients). This model is eventually simplified into a planar dynamical system describing the evolving meridional density gradients in terms of the time evolution of the meridional buoyancy gradient and buoyancy fluxes described below. This model provides a conceptual framework to explain the D-O oscillation as a noisy excitable fast-slow non-linear dynamical system. In the original model of Stommel [1], it was assumed that the ocean relaxes to a fixed atmospheric meridional temperature gradient on a much faster timescale than the salinity in the model [3]. Here we combine both temperature and salinity, represented in a buoyancy framework, on a timescale that is fast compared with a slower timescale surface buoyancy forcing. The simple model is constrained by the comprehensive model results and therefore will be parameterized to a significant extent in order to reproduce the results from the comprehensive model.

Box model framework

The fluxes of heat ($Q = [W\ m^{-2}]$) and freshwater ($F = [m\ s^{-1}]$) are surface fluxes acting over the area A of the North Atlantic Box (Figure S1). Initially, we will formulate the conservation equations for heat and salt in terms of Q and F and assume that the fluxes of heat and freshwater leaving or entering the surface of the North Atlantic box are nearly balanced by the fluxes from the lower-latitude regions [3]. This turns out to be a fair approximation if one diagnoses the fluxes in the comprehensive model. The Equatorial Atlantic atmosphere transports some freshwater to the Pacific and the remainder is balanced by runoff into the Arctic which returns to the North Atlantic as sea ice, as well as the contribution of the regular P-E in the North Atlantic. We will, however, eventually divert from this formulation to include the buoyancy fluxes from the Southern Ocean used in the production of Antarctic Bottom Water (AABW). We find that a core component of the D-O oscillation is the accumulation of ocean heat content during stadials and the subsequent release of heat to space during interstadial to maintain a long term climate balance in the top of the atmosphere radiation balance [4]. We will argue that it is this residual heat accumulation/release in the ocean that is a core component of the D-O oscillation [5]. With ocean heat accumulation or release, there must be an imbalance of buoyancy flux over the millennial timescale. Therefore, the key to the slow-fast system is to monitor the residual buoyancy imbalance in the surface of the Atlantic and assume this balances buoyancy fluxes from other parts of the globe in order to close the global overturning circulation (we will assume the majority of the balance originates in the Southern Ocean). We have divided the comprehensive model into 7 regions and represented these regions in the simple box model (Figure S1). Here the fluxes of heat (Q) and freshwater (F) per unit area are assumed to be a volume average of the fluxes entering or leaving

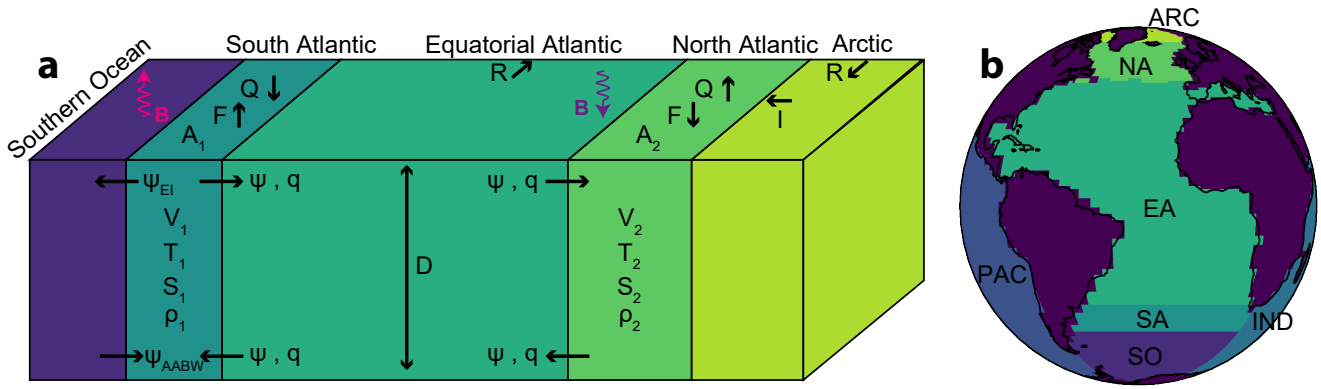

Figure S1. a) The D-O Box Model: The volume and area of the boxes are V and A, while the depth of the pycnocline is D. Temperature (T), Salinity (S) and Density (ρ) in each of the boxes are used to compute meridional gradients. The heat (Q) and freshwater (F) represent fluxes into and out of the simple box model. The Atlantic-Pacific moisture transport (R) flows out of the Equatorial box. The sea ice melt (I) (from Arctic to North Atlantic) is balanced by the runoff (R) in the Arctic Box, which balances the the meridional atmospheric moisture transport (R) from the low-latitude Atlantic to Pacific and South to North. ψ is the meridional volume transport (AMOC) and q is the freshwater transport by the AMOC. The residual buoyancy flux (B) is the long (slow) time-scale imbalance of the heat and freshwater fluxes absorbed into the Atlantic basins (SA + EA + NA). This is assumed to balance the long timescale anomalous buoyancy loss (-B) along the coast of Antarctica **b)** The glacial ocean model grid from the comprehensive climate model. The basins from the comprehensive model are used to calculate mean quantities of T, S, and ρ and correspond to each of the basins in the box model. SO = Southern Ocean, SA = South Atlantic, EA = Equatorial Atlantic, NA = North Atlantic, A = Arctic, PAC = Pacific, IND = Indian.

other parts of the system. With this assumption, and using the simplification that $A_1 = A_2 = A$ and $V_1 = V_2 = V$, we can develop a simplified set of quasi-conservative equations for heat and salt in a Stommel like system of equations [1].

We start by considering how incoming freshwater fluxes (e.g. evaporation and precipitation) change salinity in a bucket filled with salt water. Mass and salt conservation imply, respectively,

$$\begin{aligned} \frac{dV}{dt} &= FA \\ \frac{d}{dt}(VS) &= 0 \end{aligned} \tag{1}$$

Changes in salinity of an ocean water column (3.5 km) due to evaporation, precipitation, runoff and sea ice melt ($\approx$1m/year) are small and therefore we do not have to consider mass exchanges due to these processes. Under the assumption that $S = S_0 \approx 35.0$ practical salinity units (psu), we introduce a "virtual salt flux" per unit area ($S_0F : F+\uparrow$) to arrive at the approximate equations:

$$\begin{aligned} V\frac{dT}{dt} &= -T\frac{dV}{dt} = \frac{QA}{\rho_0 c_p} \\ V\frac{dS}{dt} &= -S\frac{dV}{dt} = SFA \approx S_0FA \end{aligned} \tag{2}$$

We express the conservation equations as the following set of conservation equations for temperature and salinity for the South Atlantic (Box 1) and North Atlantic (Box 2), see Figure S1:

$$\begin{aligned} V\frac{dT_1}{dt} &= \frac{QA}{\rho_0 c_p} - \psi\left(T_1 - T_2\right) \\ V\frac{dT_2}{dt} &= \frac{-QA}{\rho_0 c_p} - \psi\left(T_2 - T_1\right) \\ V\frac{dS_1}{dt} &= FAS_0 - |q|\left(S_1 - S_2\right) \\ V\frac{dS_2}{dt} &= -FAS_0 - |q|\left(S_2 - S_1\right) \end{aligned} \tag{3}$$

The surface heat fluxes ($Q: + \downarrow$) are driven by the balance between incoming and outgoing shortwave and longwave radiation, as well as sensible and latent heat fluxes. The boxes have an average glacial North Atlantic density (ρ_0) of 1028.5 $kg\ m^{-3}$ and a heat capacity (c_p) of 4000 $J\ kg^{-1}\ K^{-1}$. The total freshwater flux into the surface is composed of the evaporation, precipitation and runoff (P-E+R) into and out of the boxes acting over the area of the boxes. The mass/volume transport by the AMOC is ψ in units of $m^3 s^{-1} = 10^{-6} Sv$. The freshwater transport by the AMOC (q) is in units of Sv of freshwater. An association will later be made below between the freshwater transport by the AMOC (q) and the convergence and divergence of salt in the Atlantic between 34°S and 60°N (e.g. the transport of salt by the baroclinic meridional velocity, F_{ov} or M_{ov} [6–8]). The absolute value sign arises because the buoyancy (density) anomalies between southern and northern boxes are modified in the same way irrespective of the direction of the transport [1].

We reduced the dimensionality of the system of equations 3 further by defining the meridional variations of temperature and salinity as follows,

$$\Delta S = S_1 - S_2;\ \Delta T = T_1 - T_2;\ \Delta\rho = \rho_1 - \rho_2 \quad (4)$$

We also use a linear equation of state for sea water (LEOS),

$$\begin{aligned} \frac{\rho}{\rho_0} &= 1 + \beta_S (S - S_0) - \alpha_T (T - T_0) \\ \Delta\rho &= \rho_0 (\beta_S \Delta S - \alpha_T \Delta T) \end{aligned} \quad (5)$$

where the thermal expansion and haline contractions coefficients are (α_T) and (β_S), respectively and T_0 and S_0 are the reference temperature and salinity, respectively. Subtracting the evolution equations for temperature and salinity (equation set 3), we obtain relations for ΔS and ΔT,

$$\begin{aligned} V\frac{d}{dt}(\Delta T) &= \frac{2QA}{\rho_0 C_p} - 2\psi(\Delta T) \\ V\frac{d}{dt}(\Delta S) &= 2FS_0A - 2|q|(\Delta S) \end{aligned} \quad (6)$$

The factor of 2 arises under the assumption of a symmetry of the fluxes between the southern and northern boxes [3]. To simplify further, we write the equation for the time variation of the density difference using the linear equation of state for seawater (LEOS),

$$V\frac{d}{dt}(\Delta\rho) = V\rho_0 \frac{d}{dt}(\beta_S \Delta S - \alpha_T \Delta T) \quad (7)$$

Inserting the prognostic equations for the meridional gradients of temperature and salinity (equations 6) into the meridional density difference LEOS (equation 7) and replacing ψ with q, we obtain,

$$\begin{aligned} V\frac{d\Delta\rho}{dt} &= 2\rho_0\beta_S\left[FS_0A - 2|q|\Delta S\right] \\ &\quad - 2\rho_0\alpha_T\left[\frac{QA}{\rho_0 C_p} - 2|q|\Delta T\right] \end{aligned} \quad (8)$$

The assumption of replacing ψ with q in effect allows the freshwater transport to move heat as well as salt in the buoyancy formulation below. While this is somewhat unrealistic, it allows for the reduction in the number of degrees of freedom of the simple system. When we introduce the parameterization of the freshwater transport from the comprehensive model into the simple model below, the simple model will be constrained to the complex model behaviour. This parameterization obtained from the comprehensive model, will use a linear relation to equate the freshwater transport to the magnitude of the buoyancy (density) gradient and therefore emulate the movement of both heat and salt in the meridional transport in the simple model (see below). The general expression for the total surface buoyancy flux is a combination of the haline (F: freshwater flux) and thermal (H: heat flux) forcing given by [9],

$$B_F + B_H = -g\beta_S S_0 F + g\alpha_T \frac{Q}{\rho_0 C_p} \quad (9)$$

Here g is the force of gravity ($m\ s^{-2}$). The removal of freshwater (F +up) out of the ocean surface decreases the fluid buoyancy (ocean water becomes saltier). The input of heat (Q +down) into the ocean surface increases the fluid buoyancy. This notation is used to be consistent with the conservation equations 3. The prognostic density (equation 8) can be

transformed into a buoyancy (b) equation, where the buoyancy ($m\ s^{-2}$) is defined as,

$$b = -g\frac{(\rho - \rho_0)}{\rho_0} \tag{10}$$

and the meridional difference in buoyancy between the southern and northern box (S - N) is,

$$\begin{aligned}\Delta b &= b_1 - b_2 \\ &= -g\left(\frac{\rho_1 - \rho_0}{\rho_0}\right) + g\left(\frac{\rho_2 - \rho_0}{\rho_0}\right) \\ &= \frac{-g}{\rho_0}\Delta\rho\end{aligned} \tag{11}$$

Combining equations 8 and 11 we obtain,

$$\frac{d\Delta b}{dt} = -\frac{g}{V} \left(2\beta_S S_0 FA - 2\alpha_T \frac{QA}{\rho_0 c_p} - 2|q|\left(B_S \Delta S - \alpha_T \Delta T\right)\right) \tag{12}$$

and using the general forms for thermal and haline buoyancy flux (equation 9) we simplify as,

$$\frac{d\Delta b}{dt} = \frac{2}{D}\left(B_F + B_H\right) - 2\frac{|q|}{V}\Delta b \tag{13}$$

Here B_F and B_H denote haline (freshwater) and thermal (heat) buoyancy fluxes, respectively. The depth of the boxes (D) is approximately the pycnocline depth at 1000 meters. The assumption here is that on the fast timescale the fluxes going into the low latitude Atlantic are balanced by the fluxes going out of the North Atlantic box.

We find in our analysis of the comprehensive model that the buoyancy fluxes defined above (B_F and B_H) have a fast timescale behaviour consistent with the changes in the North Atlantic D-O oscillation. However, since the lower latitude buoyancy fluxes do not balance the high latitude buoyancy fluxes, we can examine this anomaly and attribute it to the evolution of the slow timescale buoyancy fluxes over the Atlantic sector north of the Southern Ocean. We now reformulate this simple model equation in terms of the slow timescale by instead considering a residual buoyancy flux which will be used to capture this imbalance of buoyancy in the Atlantic ocean climate system over the millennial timescale. We will define, B, as the area weighted sum of the buoyancy fluxes over the Atlantic boxes, namely the Southern Atlantic, the Equatorial Atlantic and the North Atlantic. The fluxes (F and Q) out of or into these boxes do not balance on the short timescale and therefore (B) will provide a diagnostic for the interhemispheric (bipolar seesaw) balance of buoyancy fluxes in the D-O oscillation (see B in Figure S1). We also assume that the buoyancy being removed from the Southern Ocean to produce AABW ($-B$) is balanced by the residual buoyancy fluxes over the Atlantic Basin (SA, EA, NA). This balance is demonstrated in Figure S2. Therefore, we write the fast timescale equation for the meridional buoyancy gradient as,

$$\frac{d\Delta b}{dt} = \frac{-B - (B)}{D} - 2\frac{|q|}{V}\Delta b \tag{14}$$

In place of the capillary constant usually employed to relate the meridional density gradient with the LEOS [1], we instead used a fitted (parameterized) relation from the comprehensive model as,

$$\begin{aligned}\psi &= \psi_0 + \frac{\psi_1}{\rho_0}\Delta\rho &&= \psi_0 + \frac{\psi_1}{b_c}\Delta b \\ q &= q_0 + \frac{q_1}{\rho_0}\Delta\rho &&= q_0 + \frac{q_1}{b_c}\Delta b\end{aligned} \tag{15}$$

The fidelity of the linear relation of the AMOC and freshwater transport dependence on meridional buoyancy gradient (Δb) is clearly demonstrated in Figures S2, S3 and S4.

Figure S3 demonstrates a clear linear fit of the AMOC volume transport (ψ) to the buoyancy (or density) gradient and therefore the validity of the density driven transport in the comprehensive model. The gradient is taken between the Southern Atlantic box and the North Atlantic box, but the same linear behaviour is seen if one chooses the Equatorial box instead of the Southern Atlantic box. The slopes of the linear fit are nearly the same, but do vary slightly with change in CO_2 concentration in the comprehensive model. We assume that the slope remains constant in our derivations below. The freshwater transport by the AMOC

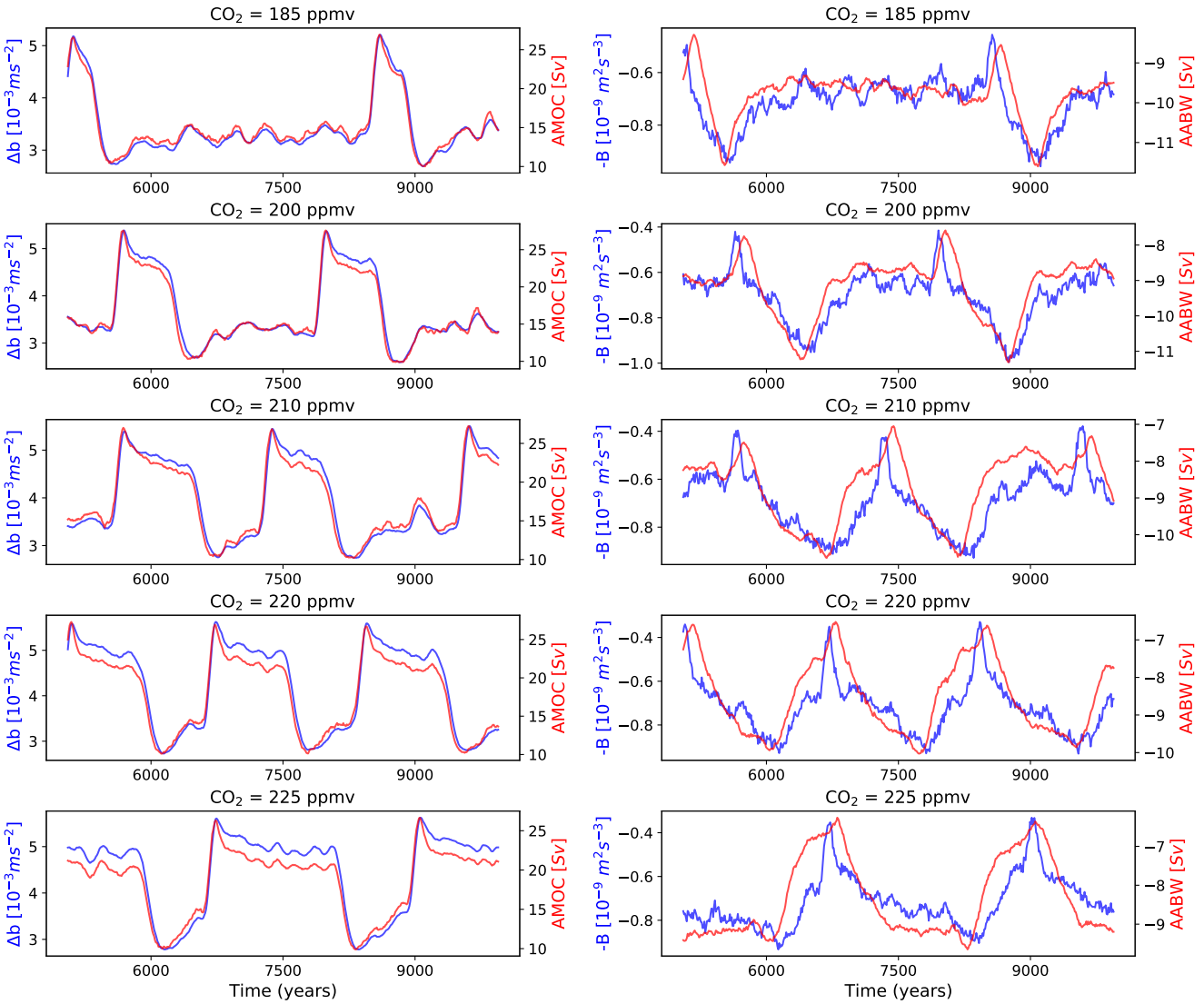

Figure S2. Left panels: The correlation between AMOC and the South Atlantic - North Atlantic meridional buoyancy gradient (Δb) in simulations with different atmospheric CO_2 concentration. Right panels: The correlation between AABW and the negative of the buoyancy flux residual ($-B$) in the Atlantic (area weighted South Atlantic + Equatorial Atlantic + North Atlantic boxes) in simulations with different atmospheric CO_2 concentration.

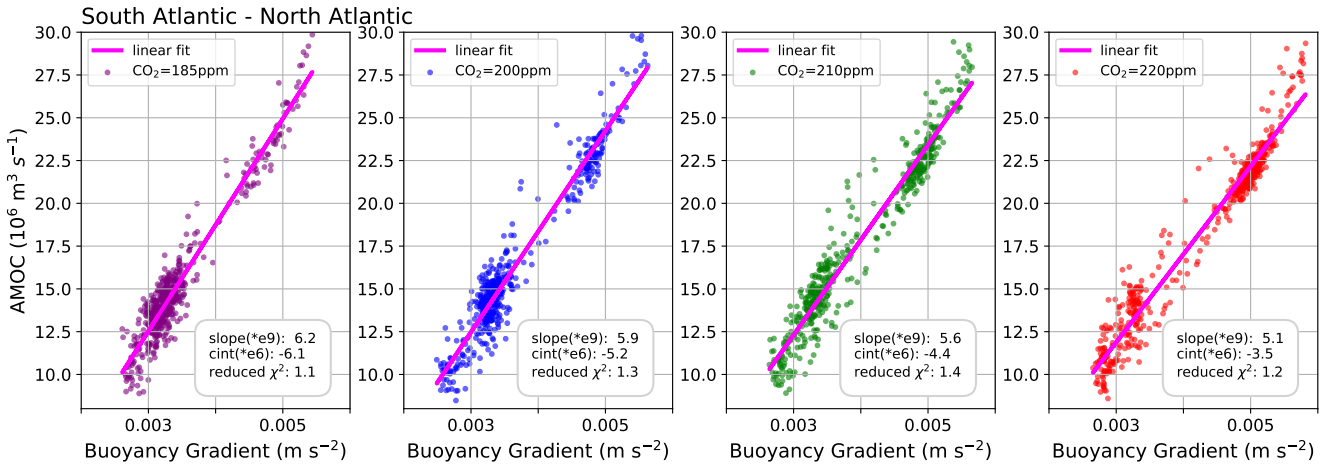

Figure S3. The linear relationship between AMOC strength and the meridional density gradient between the South Atlantic and the North Atlantic for different levels of atmospheric CO_2

(q) and the relation between freshwater convergence/divergence by the AMOC (F_{ov}) and it's implementation in a Stommel type model formulation is not obvious. However, we can use intuition to make the connection between salt transport in the simple model to a freshwater divergence/convergence term (F_{ov}) in the comprehensive model via the relation,

$$F_{ov} = -2q\frac{\Delta S}{S_0} \tag{16}$$

A similar form has been used in a previous box model study [8], but is different in this study due to the reduced degrees of freedom used in our box model. In order to visualize this relation, one has to consider the set of cases when freshwater is entering/leaving the Equatorial box in the simple model (at 34°S) and entering/leaving the Arctic box (at 60°N) as occurs in the comprehensive model. The set of six permutations on the magnitude and direction of the freshwater transport at the boundaries of a simple two box model can be related to freshwater transport between boxes. This allows one to produce a linear relation (equation 16) for the freshwater transport (in Sv) from the comprehensive model and gives rise to the hysteresis behaviour observed in the comprehensive model (i.e. the freshwater transport switches direction whereas the volume transport never does). This will have important implications for the hysteresis curve that usually follows a typical '2' shaped structure in a salinity framework [10]. Since we use equation 16 and equation 15 to relate the freshwater transport to the meridional buoyancy gradient in the comprehensive model, this has the action of transposing the typical salinity hysteresis curve because the comprehensive model is advecting positive buoyancy (heat) anomalies northward that weakens a negative meridional (S - N) buoyancy gradient (lighter more buoyant water at low latitude and heavier less buoyant at high latitude). The northward advection of positive salinity anomalies would act to enhance a negative meridional (S - N) buoyancy gradient.

Inserting the freshwater exchange function 15 into equation 14 gives,

$$\frac{d\Delta b}{dt} = -\frac{2B}{D} - \frac{2}{V}\left|q_0 + \frac{q_1}{b_c}\Delta b\right|\Delta b \tag{17}$$

where b_c is approximately the mean buoyancy in the linear density driven relations (Figure S4). If an auto-oscillatory system does indeed exist in the comprehensive model, the buoyancy flux (B) would have a slow-timescale behaviour that represents the forcing that tips the system through a set of saddle node bifurcation points determined from the steady state of the fast-timescale evolution equation 14 (the slow manifold: $d\Delta b/dt = 0$),

$$B = -\frac{D}{V}\left|q_0 + \frac{q_1}{b_c}\Delta b\right|\Delta b \tag{18}$$

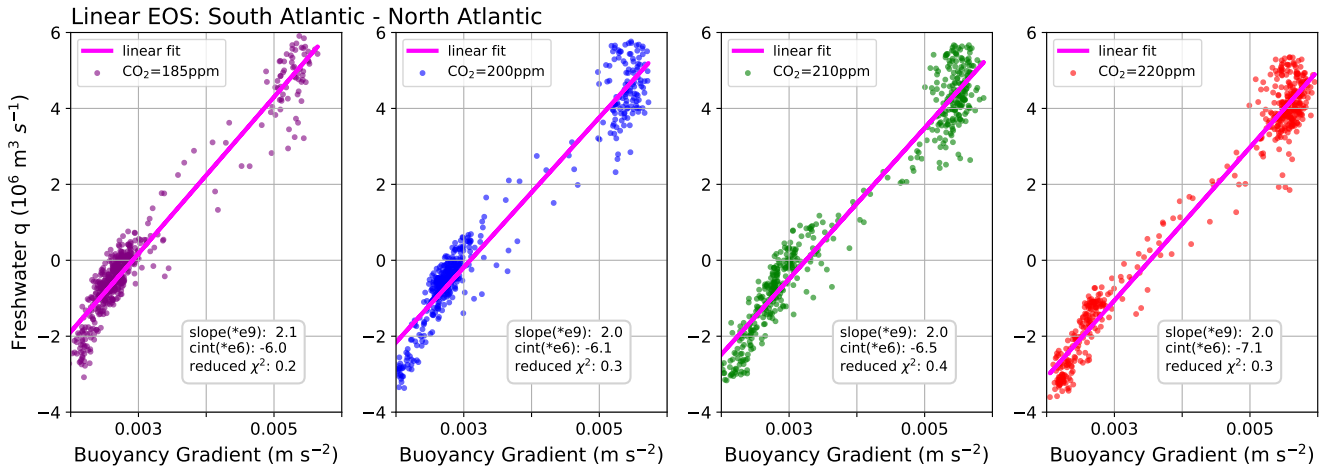

Figure S4. The linear relationship between freshwater transport by the AMOC and the meridional density gradient between the South Atlantic and the North Atlantic for different levels of atmospheric CO_2. The buoyancy gradient is calculated with a LEOS for consistency with the simple model equations. For the simple model we use the intercept and slope values of $q_0 = -6.0x10^6\ m^3\ s^{-1}$ and $q_1/b_c = 2.0x10^9\ m^2\ s$ (Table S1).

We introduce a slow-timescale evolution equation for the buoyancy flux that is inferred from the slow timescale dynamics observed in the comprehensive model (e.g. the Southern Hemisphere bipolar seesaw behaviour or changes in global average ocean temperature observed in this study). We simply use the following equation below to represent the slow timescale dynamics that employs CO_2 (γ in units of buoyancy) as a control parameter,

$$\frac{dB}{dt} = \frac{F_0}{\tau}(\Delta b - \gamma) \tag{19}$$

Alternatively we could also use,

$$\frac{dB}{dt} = \frac{\kappa_v}{D\tau}(\Delta b - \gamma) \tag{20}$$

The mechanisms responsible for the slow timescale evolving buoyancy fluxes are numerous in the comprehensive model and we must therefore choose a single relatively long timescale to emulate the slow-fast system observed in the ensemble of simulations from the comprehensive model. For simplicity we choose the slow-timescale (τ = 150 years) associated with the recharge and discharge of heat in the sub-surface North Atlantic which is represented by the average temperature in this Atlantic Box (Figure S5). Also for simplicity we preclude an implied stadial-interstadial symmetry on the timescale on which the buoyancy flux operates although this is not exactly the case as seen in Figure S5. This also correlates with the global ocean taking up heat during cold stadial periods and releasing heat during warm interstadial periods [4]. Also important is the F_0 parameter which represents a flux, which is chosen as the haline flux (= 3 $m\ yr^{-1}$) that drives the system slowly between the warm and cold climate states. When the AMOC is strong and the buoyancy gradient is large the rate of change of buoyancy flux grows which has the tendency to force a decrease in the buoyancy gradient (AMOC). Likewise, when the AMOC is weak and the buoyancy gradient is small, the rate of change of buoyancy flux decreases and has the tendency to force an increase in the the buoyancy gradient (AMOC). Therefore, the haline flux (F_0) in the slow equation (equation 19) represents sea ice melt into the North Atlantic box during the interstadial (originating in the Arctic). In the stadial, salt advection feedback and/or downward thermal diffusion into the equatorial thermocline which mechanically drives the AMOC may play a role (equation 20). Due to the reduced dimensionality of the

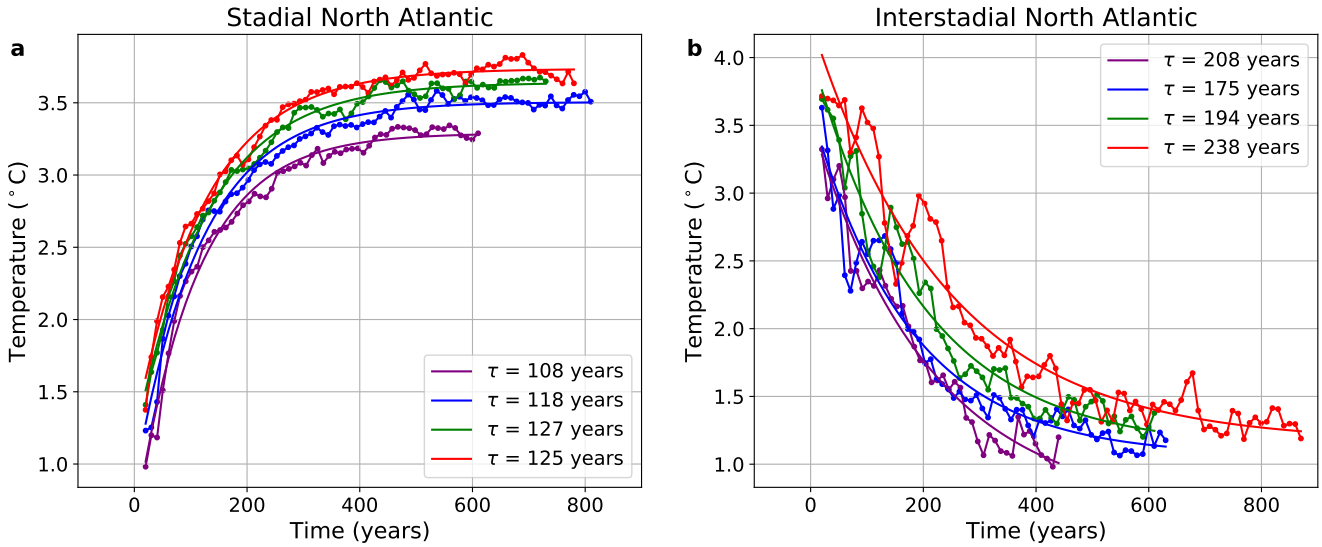

Figure S5. The recharge/discharge heat oscillator present in the North Atlantic. **a)** The stadial accumulation of heat in the North Atlantic box from the comprehensive model, which mostly occurs beneath the sea-ice covered North Atlantic. The curves have been fitted with an exponential to find a timescale (τ) associated with the accumulation of heat in four of the CO_2 experiments. **b)** Same as in **a)** but for the interstadial period. CO_2 = [185 ppm (purple), 200 ppm (blue), 210 ppm (green), 220 ppm (red)]

system, the flux (F_0) is set equal during both warm and cold periods. The tendency of the buoyancy flux is controlled by the CO_2 control parameter (γ). When CO_2 is high (γ large) the system spends more time in the upper warm branch of the slow manifold. Warmer temperatures over the North Atlantic and Arctic reduce mean sea ice volume (Figure S6) and also warm northern waters, therefore it takes more time for the glacial climate to terminate the AMOC. The reduced AMOC state during a D-O stadial period results in less transport of salty waters to high-northern latitudes of the North Atlantic. Therefore, there is a build up of salty warm water in the low-latitude Atlantic that has been observed in a number of modelling studies [4, 11, 12]. The difference observed here is that the high CO_2 experiments have increased low latitude Atlantic salinity built up in the upper ocean during the stadial relative to the ensemble mean, which increases the tendency to re-invigorate the AMOC, therefore the system spends less time in the stadial period (Figure S7). The converse is true at lower CO_2 levels where the system spends more time in the lower cold branch of the slow manifold. Higher mean levels of Arctic and North Atlantic sea ice lead to longer stadials with smaller salinity anomalies at low latitudes, salinity anomalies that are required to re-invigorate the AMOC (Figure S7).

We now express the simple slow-fast system, equations 17 and 19, in non-dimensional form ($'$) as,

$$\begin{aligned} \frac{d\Delta b'}{dt'} &= -B' - \left|q'\right|\Delta b' \\ \frac{dB'}{dt'} &= \frac{1}{\tau'}\left(\Delta b' - \gamma'\right) \\ q' &= q'_0 + q'_1 \Delta b' \end{aligned} \tag{21}$$

with the following characteristic constants (e.g. $t = t' t_c$, ...),

$$t_c = \frac{D}{2F_0}\ ;\ \ q_c = \frac{VF_0}{D}\ ;\ \ B_c = F_0 b_c\ ;\ \ b_c = \gamma_c \tag{22}$$

Furthermore, we can infer a fast advection timescale (e.g. V/q) by using the freshwater transport to formulate the system as a slow-fast system,

$$\begin{aligned} \frac{d}{dt'}\left(\Delta b'\right) &= -\frac{1}{\tau_{fast}}\left(B^* + \left|a^* + b^*\Delta b'\right|\Delta b'\right) \\ \frac{dB'}{dt'} &= \frac{1}{\tau_{slow}}\left(\Delta b' - \gamma'\right) \end{aligned} \tag{23}$$

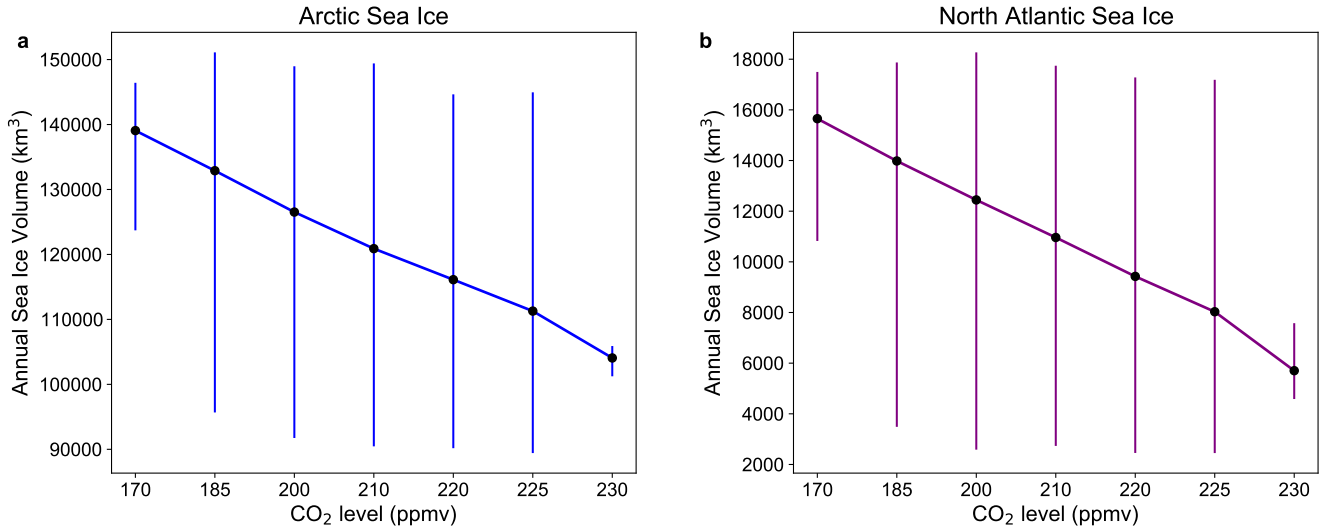

Figure S6. Sea ice volume variations in the **a)** Arctic box and the **b)** North Atlantic box for each of the different CO_2 simulations. The vertical lines span the range between minimum and maximum sea ice volume for each simulation. The black dots represent the mean volume of sea ice throughout the whole simulation. The sea ice volume follows the characteristic pattern of a system with a fold bifurcation and a control parameter (the atmospheric CO_2 concentration).

Here B^* is $B'\tau_{fast}$, and similarly relations for a^* and b^* can be obtained. This set of equations has a similar form to the standard FitzHugh-Nagumo (FHN) equations [13, 14] and produces the same dynamical system behaviour,

$$\begin{aligned} \frac{dy}{dt} &= -\frac{1}{\tau_{fast}}\left(x+a+b(y^3-y)\right) \\ \frac{dx}{dt} &= \frac{1}{\tau_{slow}}(y-\gamma) \end{aligned} \tag{24}$$

where $\tau_{slow} >> \tau_{fast}$.

The simple model that is presented in this paper is parameterized to emulate the behaviour observed in the comprehensive model, therefore some adjustment parameters are added to the set of non-dimensional equations 21. Removing the non-dimensional prime ($'$) notation for simplicity, the governing set of equations used for the simple model in Figure 2 of the main manuscript are,

$$\begin{aligned} \frac{d\Delta b}{dt} &= -(B-B_0(\gamma)) \\ &\quad -|q_0+q_1(\Delta b-b_0)|\,(\Delta b-b_0) \\ \frac{dB}{dt} &= \frac{1}{\tau}(\Delta b-\gamma) \end{aligned} \tag{25}$$

Including a Wiener processes (W) with variance ($\sigma_{\Delta b}=\sigma_B=\sigma$) we can express equation 25 as a stochastic ordinary differential equation (SODE),

$$\begin{aligned} d\Delta b &= \\ &\quad [-(B-B_0(\gamma))-|q_0+q_1(\Delta b-b_0)|\,(\Delta b-b_0)]\,dt \\ &\quad +\sigma dW \\ dB &= (\Delta b-\gamma)\,dt/\tau+\sigma dW \end{aligned} \tag{26}$$

The $B_0(\gamma)$ term in equation 25 is introduced to produce horizontal shifts in the slow manifold (i.e. the AABW magnitude), a property which is observed in the comprehensive model at different levels of CO_2. Since this does not impact the position of the system fixed point on the slow manifold, the period of the D-O oscillation or the relative stadial-interstadial duration is not modified by the introduction of this parameter. The introduction of the b_0 term is used to produce vertical shifts in the slow manifold so as to span the range of physical values of the meridional buoyancy (density) gradient in the comprehensive model. The CO_2 control parameter (γ) in the above equation is formulated in terms of buoyancy and its

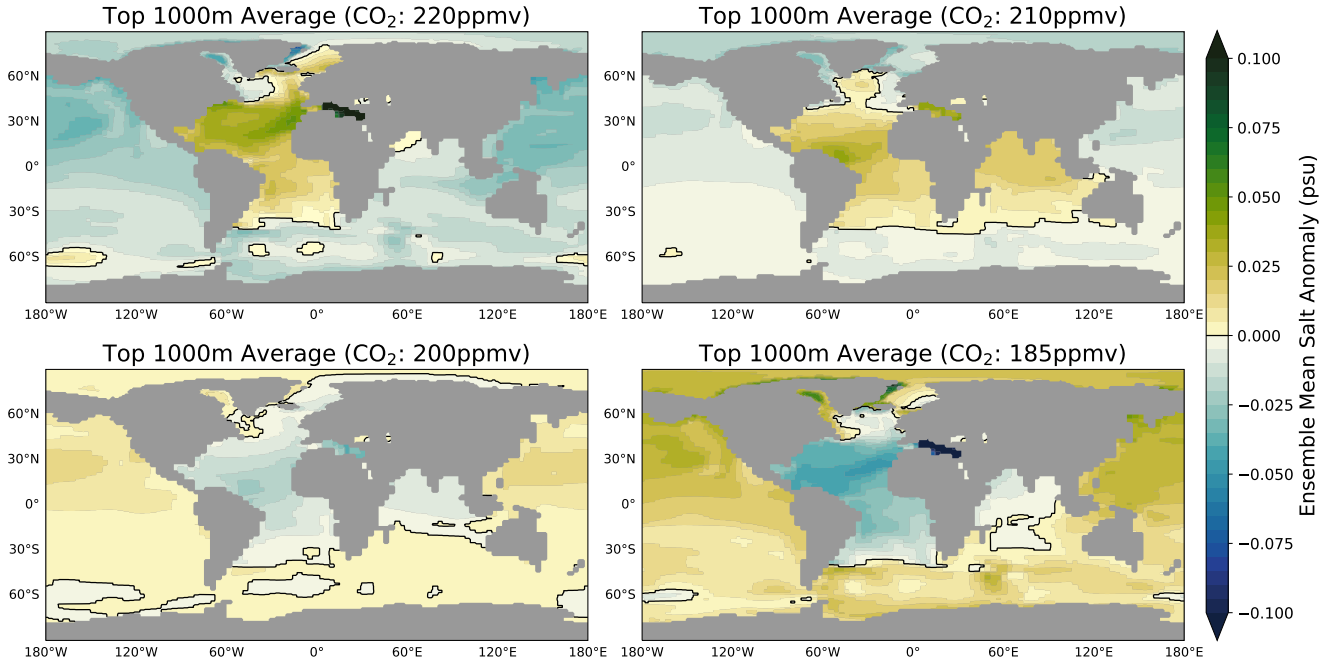

Figure S7. The stadial salinity anomaly of each the experiments with different levels of CO_2 differenced from the ensemble mean of the four experiments. The average salinity in the top 1000 meters of the ocean in each experiment is averaged and then differenced from ensemble mean average salinity in the top 1000 meters.

modifications of B_0 is parameterized as,

$$B_0(\gamma) = \bar{B} + \eta\,(\bar{\gamma} - \gamma) \tag{27}$$

The buoyancy flux correction (equation 27) is made up of, $\bar{\gamma}$, which is a buoyancy flux at a CO_2 level in the middle of the unstable manifold and, $\bar{B}$, which is the position of the upper saddle node in non-dimensional buoyancy flux units. γ is the CO_2 level expressed in non-dimensional buoyancy units (see below). η is chosen based upon the shift in mean AABW strength as a function of CO_2 level observed in the comprehensive model (but the complex and simple models do not quite have the same variation). The non-dimensional system is transformed to the AABW/AMOC slow-fast system via two transformations. The meridional buoyancy (density) gradient is used to transform from non-dimensional buoyancy to AMOC units (Sv) as follows,

$$\psi_{AMOC} = \psi_0 + \psi_1 \Delta b' \tag{28}$$

The non-dimensional buoyancy flux is converted to AABW units (Sv) as follows,

$$\psi_{AABW} = \psi_A + \chi A \frac{B_c}{b_c} B' \tag{29}$$

where χA is taken to be the area of the interface in the Atlantic between the NADW and AABW cells through which buoyancy is fluxed downwards. An additional adjustment factor to shift the hysteresis curve along the AABW axis is added as a final tuning parameter (ψ_A). Finally, the atmospheric CO_2 concentration (ppm) is simply configured according to a linear relation, that matches the fixed point position in the meridional buoyancy gradient space (Δb) with a specific level of the CO_2 concentration,

$$\gamma' = \frac{1}{\psi_1} \Big(\big((CO_2 - c_1)(a_2 - a_1)/(c_2 - c_1) + a_1 \big) - \psi_0 \Big) \tag{30}$$

The non-dimensional equations are expressed in non-dimensional buoyancy and buoyancy flux (Figure S8a, b). They can then also be transformed to

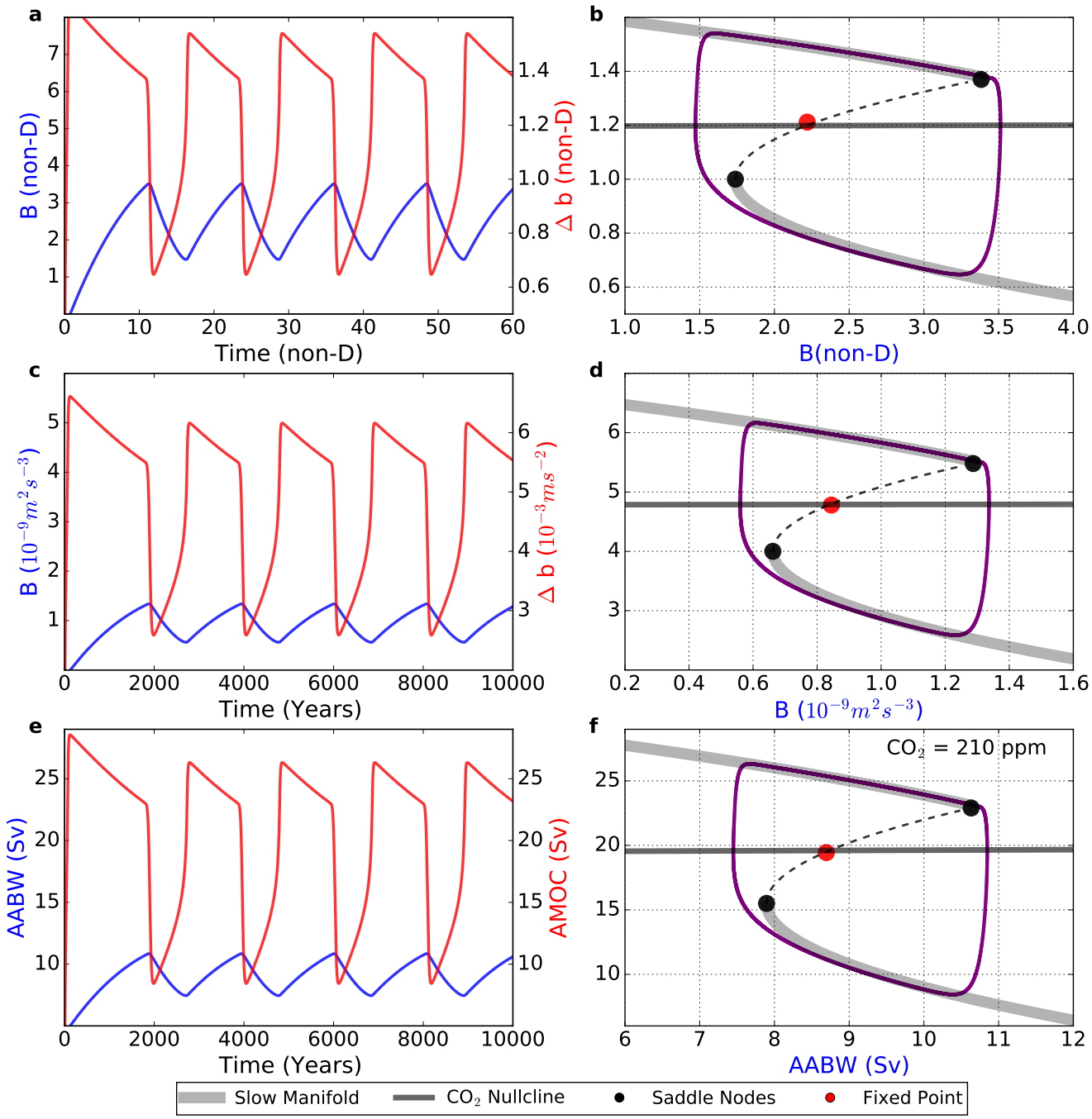

Figure S8. **a)** The simple model, with CO_2=210 ppmv, timeseries of buoyancy flux (blue) and meridional buoyancy gradient (red) in non-dimensional units and **b)** the representative phase space of buoyancy flux (B) vs meridional buoyancy gradient (Δb) colored in purple. **c)** and **d)** The same as in **a)** and **b)** but for buoyancy flux (B) vs meridional buoyancy gradient (Δb) in dimensional units. **e)** and **f)** The same as in a) and b) but for buoyancy flux (B) vs meridional buoyancy gradient (Δb) transformed to AABW and AMOC, respectively. The intersection of the slow manifold and the CO_2 nullcline cross at the system fixed point.

dimensional buoyancy and buoyancy flux (Figure S8c, d) or directly to the AMOC-AABW simple model (Figure S8e, f). Figures S8c, d are approximations representing the variations observed in the comprehensive climate model (Figure S2). The parameter estimation is obtained by correlating equations 25 and a diagnostic analysis of the comprehensive model to obtain the hysteresis diagram in Figure S9. The production of the final simple model shown in Figure S9b is derived through the set of steps illustrated in Figure S8.

The parameters used in the governing equations are given in Table S1. The parameters are based upon values obtained from the comprehensive model and are illustrated in the analysis here. The numerous detailed physics mechanisms of the D-O oscillation presented from previous analysis of the comprehensive climate model (e.g. [15]) obviously can not be introduced into the simple box model in detail. The main driver that was previously [15] found to be responsible for the slow timescale decay of the interstadial AMOC was the positive buoyancy flux associated with the melting of sea ice in the convective regions of the glacial North Atlantic. In particular, the convective activity that begins at the onset of the D-O warming event in the Greenland and Irminger Seas. The massive inventory of old Arctic sea ice that is transported through the Denmark Strait to the North Atlantic convective regions provides a continuous freshwater flux of sea ice melt while the AMOC is in a strong state of circulation. An assumption of the simple model is that the model embodies a heuristic hydrological cycle component in the sense that evaporation from the low latitude box feeds into high latitude Siberian runoff, which contributes to the formation of the large reservoir of sea ice available to the glacial North Atlantic. This interstadial negative feedback, which is taken to be dependant on the strength of the AMOC, acts to drive the system back into a reduced mode of AMOC circulation where stadial conditions are once again prevalent. The sea ice area in the Arctic and North Atlantic also appear to follow a bifurcation induced tipping behavior that is consistent with a change in the CO_2 control parameter (Figure S6).

During the stadial, a meridional haline flux [15] has been shown to destabilize the strong stably stratified halocline in the glacial North Atlantic. Salt is transported northwards into the critical convective regions of the North Atlantic in the upper halocline region. It was demonstrated in the complex climate model that the unstable vertical gradient in temperature (stable vertical gradient in salinity) is destabilized by this salt transport from low to high latitudes during the stadial in the upper layers of the North Atlantic. This is crudely represented by salt advection feedback in the simple model. The rapid increase in temperature observed at North GRIP during the transition between cold stadial conditions and warm interstadial conditions are now recognized to be associated with the release of a large reservoir of ocean available convective potential energy (OCAPE) that existed below the sea ice covered North Atlantic during stadial conditions [16, 17]. This OCAPE exists because of the strong salinity gradient which stabilizes the unstable temperature profile that exists below the extensive stadial sea ice and halocline of the North Atlantic. This sub-surface pool of warm salty water is crudely parameterized into the simple model as a thermal buoyancy flux which acts to decrease the meridional temperature gradient between north and south. We can think of the timescale associated with the build up of sub-surface heating being similar to the mechanisms associated with the charging and discharging of a capacitor in an electric circuit and this part of the system is consistent with changes in heat throughout the whole Atlantic (Figure S10) and forms an important component of the bipolar seesaw mechanism [12, 18, 19]. In the simple model we associated this recharge/discharge mechanism with the slow timescale of the system.

Other notes

Full set of glacial simulations and noise induced transitions

A number of simulations were run for the glacial climate at different levels of atmospheric CO_2 concentration and a select set was shown in the main manuscript. Figure S11 shows the full set of

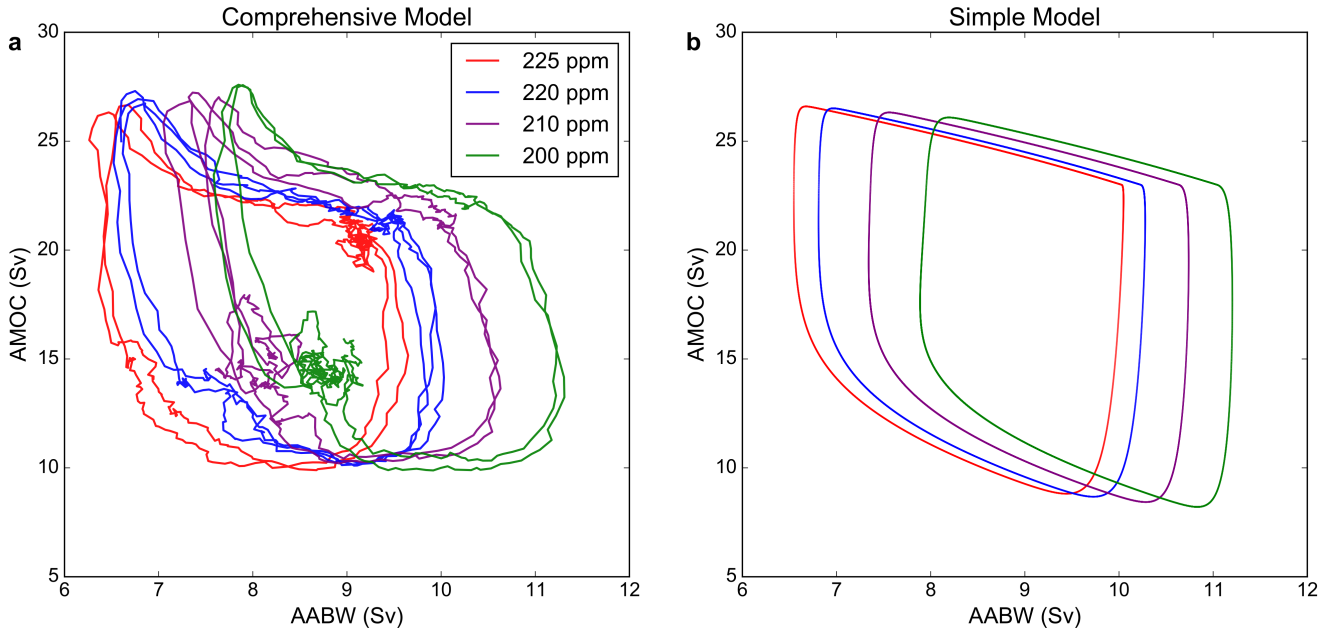

Figure S9. **a)** The phase space representation of AABW (x-axis) vs AMOC (y-axis) for four of the CO_2 experiments in the comprehensive model. The AABW has been shifted 100 years in time to remove the interhemispheric lag found between the AMOC and AABW. **b)** Same as in **a)**, but for the simple conceptual model. The simple model AABW has no time lag with the AMOC.

CCSM4 simulations that were used to measure out the D-O sweet spot. Figure S12 shows the sub-set of simulations from the main manuscript (Figure 2) without noise added and is used to demonstrate that the simple model has no oscillations at high and low levels of CO_2 concentration outside of the D-O sweet spot. The simple model with no noise illustrates how the fixed points in the comprehensive model may be very close to saddle node bifurcation points at CO_2 concentrations of 185 ppm and 230 ppm. At high CO_2 concentration early in MIS 3, small stadials may be excited by noise. At mid-level CO_2 concentration during the middle part of MIS 3 when more regularity in the D-O oscillation is observed, the system is more deterministic and noise has a smaller influence on the oscillations. Instead, the interstadial and stadial length is governed by the CO_2 control parameter. At low CO_2 concentration, late in the 100,000 year glacial cycle, the system goes back into the excitable noise regime, where D-O oscillations have short infrequent excursions to interstadial periods.

Phase space timeseries filtering

The timeseries in the model and observations used in the phase space representations from Figure 1 of the main manuscript were filtered with an empirical intrinsic mode function analysis [20]. This method decomposes the timeseries into a nearly orthogonal set of basis modes. The empirical mode decomposition (EMD) obtains a set of components that possess oscillatory features. In the case of the plain EMD algorithm, these are called Intrinsic Mode Functions (IMFs) as they are expected to have a single mode. By removing a select set of modes one can low or high pass frequency filter a timeseries. The first couple of empirical modes corresponding to the high frequencies in the timeseries were left out (This corresponds to a low pass filter with a roughly 200 year cutoff). The software used to implement the analysis in Figure 1 of the main manuscript used a python package called PyEMD. (see https://pyemd.readthedocs.io/en/latest/intro.html)

Physical Constant	Parameter	Value	Units
heat capacity of seawater	C_p	3994.0	$J\ kg^{-1}\ K^{-1}$
thermal expansion coefficient (@ 0°C, 35 psu)	α_T	5x10^{-5}	K^{-1}
haline contraction coefficient (@ 0°C, 35 psu)	β_S	0.78x10^{-3}	psu^{-1}
gravitational constant	g	9.8	$m\ s^{-2}$
reference density	ρ_0	1028.5	$kg\ m^{-3}$
reference salinity	S_0	35.0	psu
reference temperature	T_0	10.0	$°C$
pycnocline depth	D	1000.0	m
box area	A	7.0x10^{12}	m^2
box volume	V	7.0x10^{15}	m^3
buoyancy flux e-folding timescale	τ	150.0	years
characteristic time	τ_c	166.6	years
characteristic buoyancy	b_c	0.004	$m\ s^{-2}$
characteristic buoyancy flux	B_c	3.8x10^{-10}	$m^2\ s^{-3}$
characteristic CO_2 (buoyancy units)	γ_c	0.004	$m\ s^{-2}$
characteristic haline flux	F_0	3.0	$m\ yr^{-1}$
non-dimensional buoyancy flux e-folding timescale	τ	0.9	[]
characteristic volume transport	q_c	0.67	Sv
dimensional freshwater streamfunction y-intercept	q_0	-6.0	Sv
dimensional freshwater streamfunction slope (q_1/b_c)	q_1	8.0	Sv
non-dimensional freshwater streamfunction y-intercept	q_0	-9.0	[]
non-dimensional freshwater streamfunction slope	q_1	12.0	[]
buoyancy (y-axis) adjustment parameter	b_0	0.625	[]
non-dimensional mean CO_2 (buoyancy)	$\bar{\gamma}$	1.22	[]
upper saddle node buoyancy flux coordinate (B @ CO_2=210)	$\bar{B}$	3.42	[]
AMOC streamfunction y-intercept	ψ_0	-4.5	Sv
Wiener process	σ	0.06	[]
AABW streamfunction y-intercept	ψ_A	5.0	Sv
AMOC streamfunction slope (ψ_1/b_c)	ψ_1	20.0	Sv
AABW adjustment factor due to CO_2	η	0.5	[]
area multiple	χ	2.5	[]
minimum CO_2 scaling level	c_1	170.0	ppmv
maximum CO_2 scaling level	c_2	240.0	ppmv
minimum AMOC scaling level	a_1	12.0	Sv
maximum AMOC scaling level	a_2	25.0	Sv
ocean vertical mixing	κ_v	1.0x10^{-4}	$m^2\ s^{-1}$

Table S1. List of Simple Model Parameters

Software

The main software ecosystem used for the analysis in the main manuscript and supplement is python and the Anaconda python software stack. Open source contributions from the PANGEO group [21], a community platform for Big Data geoscience were also extremely helpful. The main diagnostic analysis were facilitated through xarray, dask, scipy, numpy, matplotlib, sdeint, nsim, xesmf, pop-tools and xgcm.

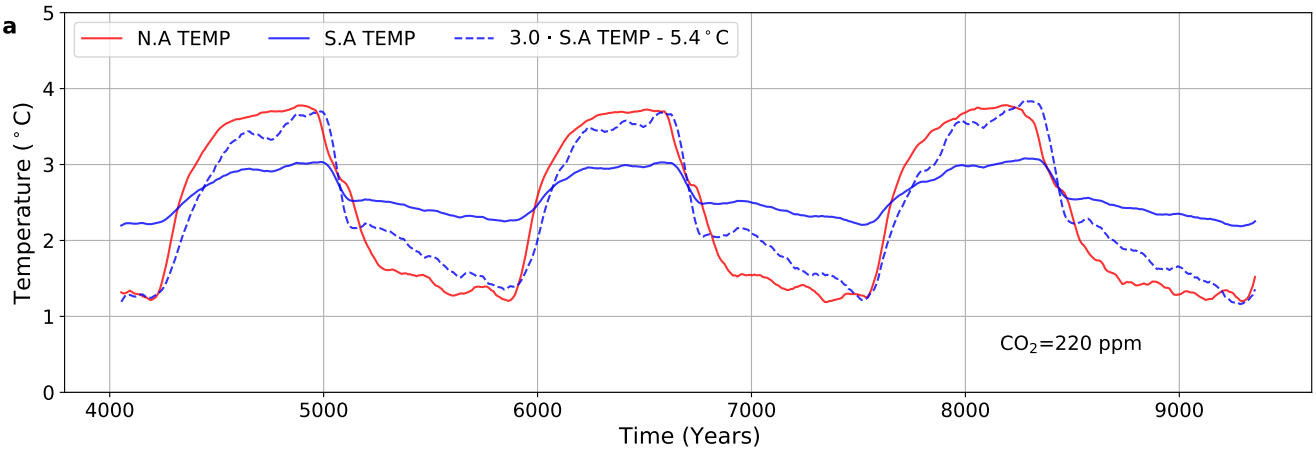

Figure S10. **a)** North Atlantic box (red) and South Atlantic box (blue) volume average temperature. The South Atlantic box is also scaled and overlapped with the North Atlantic temperature (dashed blue) showing the amplitude variations are approximately 1/3 of the changes of the North Atlantic and slightly lagged.

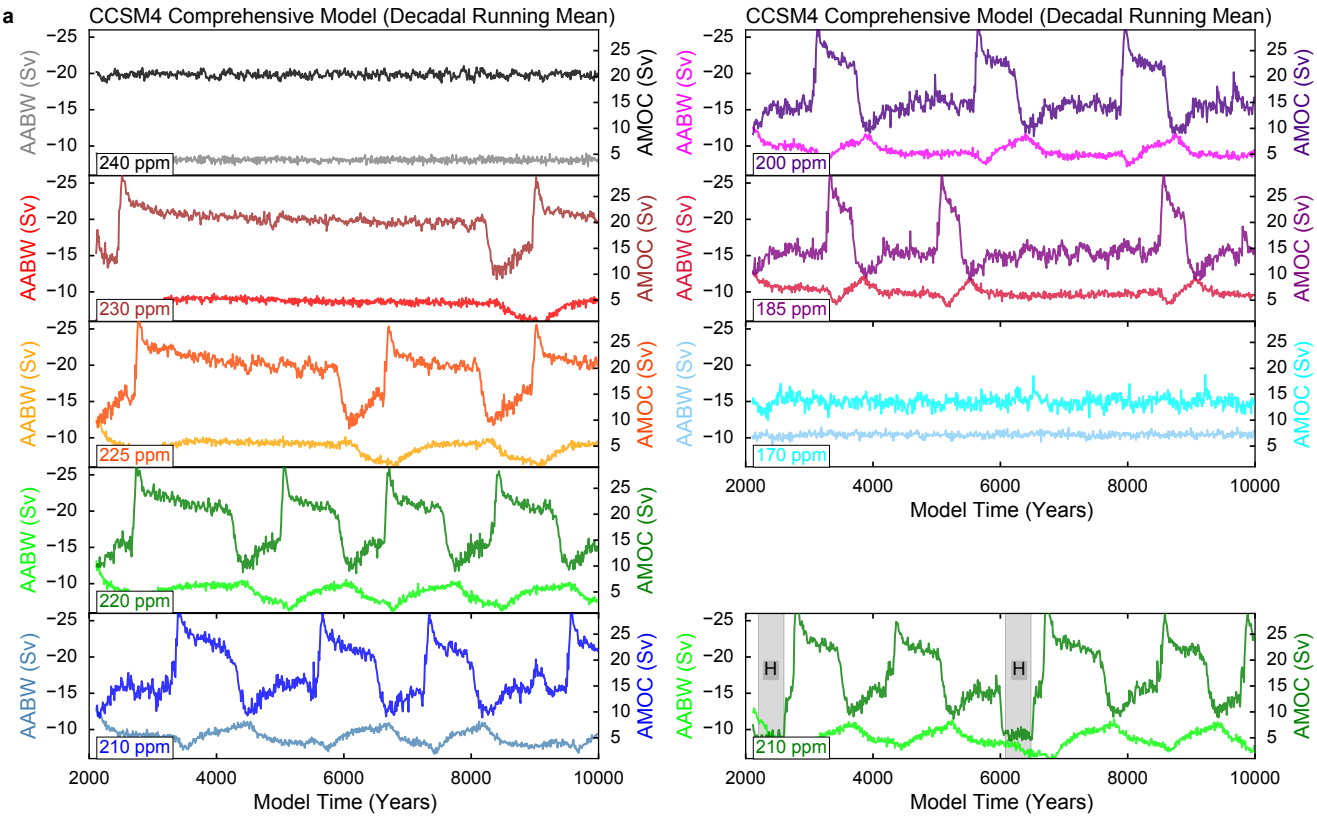

Figure S11. The set of 9 CCSM4 glacial simulations used in this study.

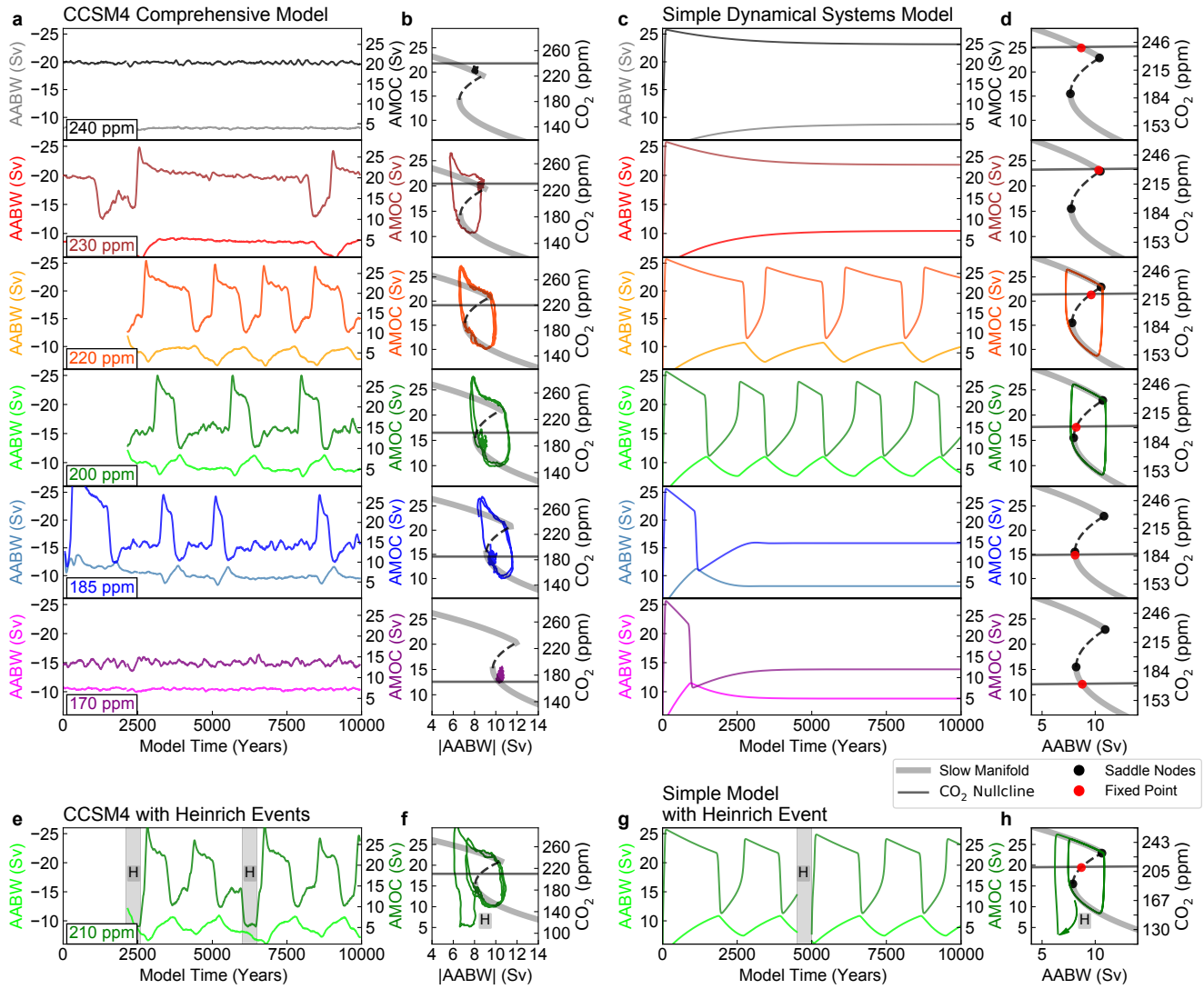

Figure S12. **a)** The set of simulations from the comprehensive model used in the main manuscript Fig. 2. **b)** The simple model without noise added to the system showing clearly at which CO_2 levels the stable and unstable solutions exist. The stable solutions near the saddle nodes (e.g. 185 ppm and 230 ppm) demonstrate how the comprehensive model may be near a saddle node bifurcation point with the D-O excursions driven by noise.

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